

An enzyme generates a Lineweaver-Burk plot with a slope of m . If the enzyme's Michaelis constant K_M is known, which expression yields the maximum velocity $V_{\max}$ of the enzyme-catalyzed reaction?

- ✔ ☒ A. K_M/m [56%]
☐ B. $K_M \times m$ [26%]
☐ C. m/K_M [13%]
☐ D. $K_M + m$ [3%]

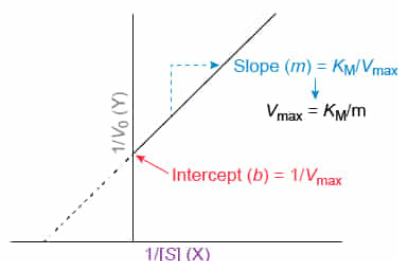
Correct
 56% answered correctly

Explanation:

Lineweaver-Burk plots

$$\frac{1}{V_0} = \frac{K_M}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}$$

$$Y = mX + b$$



Lineweaver-Burk plots are derived from a [rearrangement](#) of the Michaelis-Menten equation. They are double-reciprocal plots in which the reciprocal of reaction velocity ($1/V_0$) is plotted on the y-axis as a function of the reciprocal of substrate concentration ($1/[S]$) on the x-axis. The general form of the Lineweaver-Burk equation is

$$\frac{1}{V_0} = \frac{K_M}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}$$

This equation [converts Michaelis-Menten data](#) into a **linear plot** ($y = mx + b$) in which the **slope (m)** is $K_M/V_{\max}$ and the **y-intercept (b)** is $1/V_{\max}$. Therefore, the relationship of the slope to K_M and $V_{\max}$ can be written as

$$m = \frac{K_M}{V_{\max}}$$

If the Michaelis constant K_M is known, then the value of $V_{\max}$ can be determined by solving for it algebraically. First, both sides of the equation are multiplied by $V_{\max}$ to give

$$m \times V_{\max} = K_M$$

Then both sides are divided by m to yield $V_{\max}$ in terms of K_M and m :

$$V_{\max} = \frac{K_M}{m}$$

(Choice B) K_M should be *divided* by m , not multiplied, to yield $V_{\max}$.

(Choice C) m/K_M is the *reciprocal* of $V_{\max}$. $V_{\max}$ is equal to K_M/m .

(Choice D) K_M should be *divided* by m , not added to it, to yield $V_{\max}$.

Educational objective:

The Lineweaver-Burk equation is a rearrangement of the Michaelis-Menten equation that converts enzyme kinetic plots into linear plots. The slope of a Lineweaver-Burk plot is equal to $K_M/V_{\max}$ and the y-intercept is equal to $1/V_{\max}$.

The K_M of an enzyme is always decreased by which class of inhibitor?

- ☐ A. Mixed [1%]
☐ B. Noncompetitive [11%]
☐ C. Competitive [22%]
✔ ☒ D. Uncompetitive [64%]

Correct

64% answered correctly

Explanation:

Type of inhibition		Inhibitor binds	Effect on K_M	Effect on V_{max}
Competitive		E only	Increases	No change
Mixed (3 versions)	Favors free enzyme	E more than ES	Increases	Decreases
	Noncompetitive (special case of mixed)	E and ES equally	No change	Decreases
	Favors ES complex	ES more than E	Decreases	Decreases
Uncompetitive		ES only	Decreases	Decreases by same factor as K_M

E = free enzyme; ES = enzyme-substrate complex.

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Enzyme inhibitors are classified according to the mechanism by which they interfere with enzyme activity. **Competitive inhibitors** bind the **free enzyme E** exclusively and prevent substrate from binding. **Uncompetitive inhibitors** bind the **enzyme-substrate complex ES** exclusively, preventing the enzyme from converting substrate to product. **Mixed inhibitors** have characteristics of both competitive and uncompetitive inhibitors. All mixed inhibitors **bind both E and ES** but may favor one over the other. When a mixed inhibitor binds **E and ES with equal affinity**, it is a **noncompetitive inhibitor**.

Different inhibitory mechanisms have different effects on an enzyme's properties. Specifically, inhibitor binding to E prevents substrate binding, so a higher substrate concentration is required to reach $\frac{1}{2}V_{max}$. This substrate concentration is the K_M of the enzyme, so **competitive inhibitors** always cause an **increased K_M (Choice C)**. Conversely, inhibitor binding to ES effectively decreases the amount of ES in solution and shifts the equilibrium toward more ES formation. This causes a decrease in the amount of substrate needed to achieve $\frac{1}{2}V_{max}$ (although V_{max} itself is also decreased) and therefore **decreases K_M** . Because **uncompetitive inhibitors** bind ES exclusively, they *always* cause a **decrease in K_M** .

(Choices A and B) Mixed inhibitors may cause a decrease, increase, or no change in K_M , depending on whether they favor binding to E (increased K_M) or ES (decreased K_M). If the mixed inhibitor has equal affinity for E and ES, it is noncompetitive and does not alter K_M .

Educational objective:

Enzyme inhibitors are classified by their mechanism of action. Inhibitors that bind free enzyme E exclusively are competitive inhibitors and cause an *increased* K_M . In contrast, inhibitors that bind enzyme-substrate ES complex exclusively are uncompetitive and cause a *decreased* K_M . Mixed inhibitors have characteristics of both uncompetitive and competitive inhibitors, and may cause an increase, decrease, or no change (ie, noncompetitive inhibitors) in K_M , depending on whether they favor binding to E or ES.

Time spent:0 sec
Subject:
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System:
Enzymes